RAMANUJAN'S PRIME-COUNTING INEQUALITY HOLDS FOR ALL $x \geq 38\,358\,837\,683$

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ABSTRACT. Ramanujan observed that $\pi(x)^2 < \frac{ex}{\log x}\,\pi\left(\frac{x}{e}\right)$ for all sufficiently large x. Dudek and Platt showed, assuming the Riemann hypothesis, that the last counterexample is $x = 38\,358\,837\,682$. Unconditionally the inequality was known for $38\,358\,837\,683 \leq x \leq e^{103}$ (Johnston) and for $x \geq e^{3158.442}$ (Axler). We close the gap: the inequality holds unconditionally for every $x \geq 38\,358\,837\,683$. The proof uses no verification of the Riemann hypothesis beyond the height $3 \cdot 10^{12}$ established by Platt and Trudgian. It combines a smoothed explicit formula of Büthe's type with three new explicit inputs:

- a log-free zero-density estimate near the 1-line, obtained by reworking Bellotti's argument with a sharp asymptotic evaluation of the Barban–Vehov mean square;
- a height-dependent zero-free region, obtained by evaluating the inequality of Bellotti, Trudgian and Yang exactly;
- a repulsion lemma showing that two hypothetical zeros close to the 1-line cannot lie close to each other in height.

Every numerical step is certified in rigorous interval (ball) arithmetic. Along the way we record several corrections to the explicit inputs we use, none of which affects the published main results.

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1. Introduction

In his second notebook (see [6], p. 112) Ramanujan stated that

$$\pi(x)^2 < \frac{ex}{\log x}\,\pi\Big(\frac{x}{e}\Big) \tag{1.1}$$

holds for all sufficiently large x, where $\pi(x)$ is the number of primes up to x. Expanding π by the prime number theorem with the error term $O(x\log^{-6}x)$ shows that the two sides differ by $\frac{x^2}{\log^6 x}\,(1+o(1))$, so (1.1) is a statement about the sixth term of the asymptotic expansion. It requires the relative error $|\pi(x)-\mathrm{li}(x)|\log x/x$ to be smaller than roughly $1/(3\log^4 x)$. Following Axler [3], let N_R denote the smallest positive integer such that (1.1) holds for every real $x \geq N_R$.

What was known.

- Under the Riemann hypothesis, Dudek and Platt [12] showed $N_R = 38\,358\,837\,683$. They combined Hassani's conditional bound [13] with a computation for $x \leq 1.15\cdot 10^{16}$. The largest prime counterexample, $38\,358\,837\,677$, had been found by Galway (see [12]).
- Unconditionally, the inequality is known in two ranges:
  - for $38\,358\,837\,683 \leq x \leq 10^{19}$ (Axler [2], building on Büthe's bounds [8]), extended to $x \leq e^{58}$ by Platt and Trudgian [23] and to $x \leq e^{103}$ by Johnston [17, Thm. 5.1];
  - for $x \geq e^{X_0}$, where X_0 fell successively as follows (see [3] for the history):

$$9658 \to 9032\ [2] \to 3915\ [23] \to 3604\ [9] \to 3361\ [18] \to 3158.442\ [3].$$

The range $e^{103} < x < e^{3158.442}$ remained open. Two recent preprints [10, 11] claim to settle it; we explain in §8.2 why their arguments are not proofs.

Result.

Theorem 1.1. *Inequality (1.1) holds for every real x with $e^{58} \leq x \leq e^{3159}$. Consequently it holds for every $x \geq 38\,358\,837\,683$, and $N_R = 38\,358\,837\,683$ unconditionally.*

The second statement follows from the first together with [17, Thm. 5.1] and [3, Thm. 1.1]. The value of N_R also uses the computations of Galway and of Dudek–Platt [12] for $x \leq 10^{11}$.

The only information about zeros of ζ that we take from a large-scale computation is the theorem of Platt and Trudgian [24]: every non-trivial zero ρ of ζ with $0 < \mathrm{Im}\,\rho \leq H_0 := 3\cdot 10^{12}$ satisfies $\mathrm{Re}\,\rho = 1/2$. All remaining information about zeros comes from explicit theorems, and our own computations are certified with rigorous ball arithmetic [1].

Method. Write $L = \log x$.

(1) By an elementary argument (§2), (1.1) at x follows from $|\pi(x) - \mathrm{li}(x)| \le \varepsilon_R(L)\, x/L$ and $|\pi(x/e) - \mathrm{li}(x/e)| \le \varepsilon_R(L)\,(x/e)/(L-1)$. Here $\varepsilon_R(L) \approx 1/(3L^4)$ is an explicit function (Definition 2.1). This in turn follows from relative bounds for $\psi(t) - t$ on $[e^{43}, x]$ (Lemma 2.4).

(2) The bound for ψ comes from a smoothed explicit formula with Logan's kernel, as in Büthe [7, 8] (§3). The zeros with $|\gamma| \le H_0$ lie on the critical line. Zeros above H_0 are controlled by three ingredients:
   (a) a zero-free region $\sigma > 1 - \lambda^*(\log t)/\log t$ (§4);
   (b) explicit zero-density estimates $N(\sigma, T)$ (§6);
   (c) a repulsion lemma for pairs of zeros close to the 1-line (§5), which feeds into (b).

(3) The binding range is $L \approx 900$, where the kernel has bandwidth $T_1 \approx e^{30}$. There the decisive zeros are those with $1 - \beta \approx 0.21/\log\gamma$, at heights just above H_0. With only the published zero-free region and density constants, the zeros above H_0 alone would exceed the available error budget several times over in this range.

New ingredients.

(1) **Zero density** (Theorem 6.1). We rework Bellotti's log-free zero-density estimate [4] for σ near 1 and $3\cdot 10^{12} < T \le e^{40}$. The main gain comes from an asymptotically sharp, fully explicit evaluation of the Barban–Vehov mean square $\sum_n \Psi(n)^2 e^{-n/X}/n$ (Theorem 6.5). This is the quantity Bellotti calls d_4; our bound is about 1.5 where her Table 5 gives 85–105. The bound uses a smoothed contour argument with a certified enclosure of $1/\zeta$ in a box. Further, smaller improvements:
   - a corrected and sharpened bound for the off-diagonal pair sum, using a crowding argument;
   - a log-free pointwise bound for the mollifier;
   - a sharper treatment of the contour integral of $\zeta(1/2 + it)$;
   - parameters optimised separately for each (σ, T)-cell.

(2) **A height-dependent zero-free region** (Theorem 4.10). Bellotti, Trudgian and Yang [5] proved $\zeta(\sigma + it) \ne 0$ for $\sigma > 1 - 1/(4.896\log t)$. We evaluate their basic inequality exactly at every height and iterate. This gives $(1-\beta)\log\gamma \ge 0.21384$ for $\log\gamma \le 29$, which is $R = 4.676$ in place of 4.896. Several of their constants are tied to $\sigma \ge 0.99346$, so we re-derive all of them for the smaller σ and the larger η we need.

(3) **Repulsion** (Theorem 5.2). Keeping a second zero, together with its reflection, in the zero sum of the same inequality shows the following. Two zeros $\beta_j + i\gamma_j$ with $(1-\beta_j)\log\gamma_j \le 0.35$ at heights near e^{29} satisfy $|\gamma_0 - \gamma_1| \ge 0.058$. This removes the near-diagonal part of the pair sum in the density estimate.

Certification. The zero-free and repulsion certificates, the constants of [5] that we re-derive, the box enclosure of $1/\zeta$, the reduction of §2 and the smoothing constant are certified with ball arithmetic (`python-flint`/Arb [1]). The final scan over L, which evaluates explicit upper sums of monotone majorants, runs in double precision; §7 explains why its rounding errors are far below its margin. The code, tables and logs are available at `https://github.com/jaideepsaipadhi/ramanujans-inequality`. Section 7 lists the certificate for each lemma.

Corrections to the literature. We found the following points while re-deriving the explicit inputs we use; none of them affects the main published results.

- In [5]:
  - a normalisation slip in the final inequality: the Lemma-10 term should be multiplied by $\kappa = 433/859$;

  - an incorrect value in their Table 1, at $p = 5$;
  - a digamma majorant whose integral is larger than stated;
  - a positivity test that is valid only for $\sigma \geq 0.99346$.
- In Büthe's smoothing estimate [7, Prop. 4] (quoted as Prop. 1 in [8]), the asymptotic quoted for ν_c does not match the stated normalisation of the kernel; our own proof of the smoothing step (Lemma 3.1) avoids the issue.

Details are in §8.

2. Reduction to bounds for ψ

Definition 2.1. For $L > 1$ and $x = e^L$, put

$$g(L) = \frac{\operatorname{li}(x)^2 - \frac{ex}{L}\operatorname{li}(x/e)}{(x/L)^2}, \qquad b(L) = \frac{2\operatorname{li}(x)}{x/L} + \frac{L}{L-1}.$$

Define $\varepsilon_R(L) := \big(-b + \sqrt{b^2 - 4g}\big)/2$ whenever $g(L) < 0$.

Asymptotically $\varepsilon_R(L) = \frac{1}{3L^4}\big(1 + O(L^{-1})\big)$; for example $\varepsilon_R(L) \cdot 3L^4$ equals 1.2706 at $L = 58$, 1.0145 at $L = 907$ and 1.0041 at $L = 3159$. Computing ε_R directly is delicate, because g is the difference of two quantities of size 1 that agree to within L^{-4}.

Lemma 2.2. *Put $A(L) = Le^{-L}\operatorname{li}(e^L)$, $u = 1/L$ and $P(u) = \sum_{k<16} k!\,u^k$. Then $g = A(L)^2 - \frac{L}{L-1}A(L-1)$, $b = 2A(L) + \frac{L}{L-1}$, and $\varepsilon_R = -2g/(b + \sqrt{b^2 - 4g})$. Moreover $A(L) = P(u) + R(L)$, where*

$$-6e^{-L/2} \leq R(L) \leq \frac{16!\,u^{16}}{1 - 34u} + 6e^{-L/2} \qquad (L \geq 57).$$

For every cell $[a, a + 0.05] \subset [58, 3160.1]$ we have $g < 0$ on the cell, and ε_R admits an explicit certified lower bound there.

Proof. The formulas for g and b follow by dividing the definitions by $(x/L)^2$ and x/L. For R, integrate $\int_a^x dt/\log t$ by parts 16 times with $a = \sqrt{x}$. The boundary term at a is $\operatorname{li}(a) - aP(2/L)/(L/2)$, which has absolute value at most $3a/(L/2)$ because A and P are both at most 1.1 for arguments ≥ 28.5. The integral remainder satisfies $0 \leq (L/x)\,16!\int_a^x dt/\log^{17} t \leq 16!\,u^{16}/(1 - 34u)$. To see this, note that the logarithmic derivative of $e^s s^{-17}$ is at least $1 - 34/L$ on $[L/2, L]$.

The cancelling part $S(u) = P(u)^2 - P\big(\frac{u}{1-u}\big)/(1-u)$ is a power series in u with integer coefficients. These are computed exactly up to order 70, and $s_0 = \cdots = s_3 = 0$, $s_4 = -1$, $s_5 = -14$. The tail is bounded by $\big(\sum_{k<16} k!\big)(2u)^{71}/(1 - 2u)$. Evaluating S as $u^4 \cdot (S/u^4)$ in ball arithmetic on each cell avoids the cancellation, and the remaining terms involve R only. This is carried out by `certify/epsR_cert.py` at 200 bits. Point values agree with a 60-digit direct evaluation; the enclosures have relative width at most $1.9 \cdot 10^{-3}$, which is the variation of ε_R across a cell. □

Lemma 2.3. *Let $L = \log x \geq 58$ and $\varepsilon = \varepsilon_R(L)$. Assume $g(L) < 0$ and*

$$|\pi(x) - \operatorname{li}(x)| \leq \varepsilon\,\frac{x}{L}, \qquad \Big|\pi\Big(\frac{x}{e}\Big) - \operatorname{li}\Big(\frac{x}{e}\Big)\Big| \leq \varepsilon\,\frac{x/e}{L-1}, \tag{2.1}$$

with at least one of the two inequalities strict. Then (1.1) holds at x.

Proof. Since $\operatorname{li}(x) > \varepsilon x/L$, (2.1) gives $0 < \pi(x) \leq \operatorname{li}(x) + \varepsilon x/L$ and $\pi(x/e) \geq \operatorname{li}(x/e) - \varepsilon x/(e(L-1))$. It therefore suffices to show $(\operatorname{li}(x) + \varepsilon x/L)^2 < \frac{ex}{L}\big(\operatorname{li}(x/e) - \varepsilon\frac{x}{e(L-1)}\big)$. After dividing by $(x/L)^2$ this reads $g + b\varepsilon + \varepsilon^2 < 0$, and with $\varepsilon = \varepsilon_R(L)$ the left side equals 0. Strictness in (2.1) makes the inequality strict. □

Writing $y = x/e$, so that $\log y = L - 1$, the hypotheses (2.1) for all x with $\log x \in [58, 3159]$ become the following two conditions:

$$\begin{aligned} |\pi(y) - \mathrm{li}(y)| &< \varepsilon_R(\log y + 1)\,\frac{y}{\log y} \quad (57 \le \log y \le 3158), \\ |\pi(y) - \mathrm{li}(y)| &< \varepsilon_R(\log y)\,\frac{y}{\log y} \qquad (58 \le \log y \le 3159). \end{aligned} \tag{2.2}$$

No monotonicity of ε_R is needed: both conditions are checked directly, cell by cell (Proposition 2.5).

Lemma 2.4 (transfer)**.** *Let $y \ge 2$. Then*

$$\pi(y) - \mathrm{li}(y) = \frac{\vartheta(y) - y}{\log y} + \int_2^y \frac{\vartheta(t) - t}{t \log^2 t}\,dt + \frac{2}{\log 2} - \mathrm{li}(2), \qquad \Big|\frac{2}{\log 2} - \mathrm{li}(2)\Big| < 1.84.$$

Moreover $0 \le \psi(t) - \vartheta(t) \le 1.4262\sqrt{t}$ for $t > 0$ [27, Thm. 13].

Proof. Partial summation gives $\pi(y) = \vartheta(y)/\log y + \int_2^y \vartheta(t)\,dt/(t\log^2 t)$. Integration by parts gives $\mathrm{li}(y) = y/\log y + \int_2^y dt/\log^2 t + \mathrm{li}(2) - 2/\log 2$. Subtracting the two identities proves the formula. □

Proposition 2.5. *Suppose that for every certified interval $[L_1, L_2] \subset [43, 3159]$ of the final scan (§7)*

$$|\psi(t) - t| \le q_{[L_1,L_2]}\,\iota_{[L_1,L_2]}\,\frac{\varepsilon_R(L_2+1)}{1 + 2/L_1}\,t + a(t) \qquad (\log t \in [L_1, L_2]). \tag{2.3}$$

Here $q_{[L_1,L_2]}$ is the ratio computed by the scan for that interval, $\iota_{[L_1,L_2]} \le 1+10^{-3}$ is the correction factor of Lemma 2.7, and $a(t) = 2 + e^{2\cdot 10^{-4}}(\log t + 10^{-4})\big(2.01\cdot 10^{-4}\sqrt{t} + \frac{1}{2}\log\log(2t^2)\big)$ collects the additive terms of Lemma 3.1 and Proposition 3.3. Suppose also that $|\psi(t) - t| \le 0.94\sqrt{t}$ for $11 < t \le 10^{19}$ [8, Thm. 2]*. Then (2.2) holds.*

Proof. Combine Lemma 2.4 with (2.3), the bound $|\vartheta - t| \le |\psi - t| + 1.4262\sqrt{t}$, and the trivial bound $|\vartheta(t) - t| \le 11$ for $t \le 11$. The integral over $[2, e^{43}]$ is at most $\big((0.94 + 1.4262)\cdot 2e^{21.5}/\log^2 2\big) + 121$.

On $[e^{43}, y]$ we use upper Riemann sums in $u = \log t$ with step $h = 0.05$. On each cell we use the largest certified bound meeting it, and the inequality $\int_a^b e^u u^{-2}du \le (e^b - e^a)/a^2$. For y in a cell $[a, a+h]$ with $a \ge 57$, the numerator is bounded over the whole cell. It is compared with the certified lower bounds of Lemma 2.2 for ε_R on $[a+1, a+1+h]$ and on $[a, a+h]$; the latter comparison is needed only when $a \ge 58$. The script `certify/transfer_check.py` carries this out in 40-digit arithmetic for every cell of $[57, 3159)$. The largest value of

$$\frac{\text{bound for } |\pi(y) - \mathrm{li}(y)|}{\min(\varepsilon_R(\log y + 1),\, \varepsilon_R(\log y)) \cdot y/\log y}$$

that it finds is 0.99577, attained at $\log y = 917$. □

Remark 2.6. The factor $(1 + 2/L_1)^{-1}$ in (2.3) is what the integral term costs. Heuristically, $\int^y \varepsilon_R(\log t + 1)\,dt/\log^2 t \approx \varepsilon_R(\log y + 1)\frac{y}{\log^2 y}(1 + 6/\log y)$, so the total is $\approx (1 + 1/L + 6/L^2) \le 1 + 2/L$. Proposition 2.5 does not rely on this heuristic; it verifies the inequality numerically, cell by cell.

Lemma 2.7. *In the scan, ε is defined by three steps of the iteration $\varepsilon_{n+1} = C/\kappa(\varepsilon_n)$, where $\kappa(\varepsilon) = (L_2 + \lambda_s)/(L_1 - \lambda_s)$, $\lambda_s = \log(10\sqrt{c}/\varepsilon)$, and $C = 0.7946^{-1}\,1.02^{-1} f\tau_{12}\sqrt{c}$. The smoothing term it charges is $f\tau_{12}$, whereas Proposition 3.6 requires $f\tau_{12}\,\kappa(\varepsilon_3)/\kappa(\varepsilon_2)$. Let $\theta = (L_1 +$*

$L_2)/((L_2+\lambda_{\max})(L_1-\lambda_{\max}))$ *and* $\kappa_0=(L_2+\lambda_{\max})/(L_1-\lambda_{\max})$, *where* $\lambda_{\max}=\log(10\sqrt{110}/\varepsilon_{\min})$ *and* $\varepsilon_{\min}=0.3\,\tau_{12}\sqrt{30}/(0.7946\cdot 1.02\cdot 10)$. *Then*

$$\kappa(\varepsilon_3)/\kappa(\varepsilon_2)\le \iota:=(1+10^{-8})\exp(\theta^3\log\kappa_0).$$

Proof. With $y_n=\log\varepsilon_n$, we have $y_{n+1}=\log C-\log\kappa(e^{y_n})$. Moreover $|\frac{d}{dy}\log\kappa|=(L_1+L_2)/((L_2+\lambda_s)(L_1-\lambda_s))\le\theta$, since $\lambda_s\le\lambda_{\max}$. Hence $|y_{n+1}-y_n|\le\theta^n|y_1-y_0|$, with $y_0=\log C$ and $|y_1-y_0|=\log\kappa(\varepsilon_0)\le\log\kappa_0$. Therefore $|\log\kappa(\varepsilon_3)-\log\kappa(\varepsilon_2)|\le\theta|y_3-y_2|\le\theta^3\log\kappa_0$. The factor $1+10^{-8}$ absorbs the Logan factor $e^{\varepsilon(c+1)/6}$ of Lemma 3.4, which is $\le 1+5\cdot10^{-10}$ whenever $T_1>H_0$. The script `certify/transfer_check.py` evaluates ι for every interval and checks $\kappa_0\le 10$ and $L_1-\lambda_{\max}>1$. At $L_1=43$ it gives $\iota\le 1+10^{-3}$, and $\iota-1$ is far smaller for larger L_1. □

3. The smoothed explicit formula

We use the smoothing of Büthe [7], with our own proof of the smoothing step (Lemma 3.1). For $c>0$ and $0<\varepsilon\le 10^{-4}$, put

$$\eta_{c,\varepsilon}(\tau)=\frac{c}{\varepsilon\sinh c}\,I_0\Big(c\sqrt{1-(\tau/\varepsilon)^2}\Big)\mathbf{1}_{|\tau|\le\varepsilon},\qquad \ell_{c,\varepsilon}(z)=\frac{c}{\sinh c}\,\frac{\sinh\sqrt{c^2-(\varepsilon z)^2}}{\sqrt{c^2-(\varepsilon z)^2}}.$$

Here I_0 is the modified Bessel function. Then $\int\eta_{c,\varepsilon}=2$, and the Fourier transform satisfies $\int\eta_{c,\varepsilon}(\tau)e^{-iz\tau}d\tau=2\ell_{c,\varepsilon}(z)$. Logan's function $\ell_{c,\varepsilon}$ is entire and even, and $\ell_{c,\varepsilon}(0)=1$. Let $\lambda=\int\eta_{c,\varepsilon}(\tau)e^{\tau/2}d\tau=2\ell_{c,\varepsilon}(i/2)\ge 2$, and let

$$\phi(\tau)=\eta_{c,\varepsilon}(\tau)e^{\tau/2}/\lambda,\qquad \psi_{c,\varepsilon}(x)=\int\psi_0(xe^{-\tau})\,\phi(\tau)\,d\tau, \tag{3.1}$$

where ψ_0 is the normalised Chebyshev function. Thus ϕ is a probability density on $[-\varepsilon,\varepsilon]$, and $\psi_{c,\varepsilon}(x)$ is a weighted average of ψ_0 over $[xe^{-\varepsilon},xe^{\varepsilon}]$. It coincides with Büthe's modified Chebyshev function [8, (3.2)–(3.4)].

3.1. **Smoothing.**

Lemma 3.1. *Let* $x\ge 10^{18}$, *let* $c\in\{30,31,\dots,110\}$ *be an integer,* $0<\varepsilon\le 10^{-4}$, *and* $\lambda_s=\log(10\sqrt{c}/\varepsilon)<\log x-1$. *Put* $\nu_c:=\int_0^1 u\,\eta_{c,1}(u)\,du$. *Then*

$$|\psi(x)-\psi_{c,\varepsilon}(x)|\le\frac{\log x+\lambda_s}{\log x-\lambda_s}\cdot 2\varepsilon\,e^{\varepsilon/2}\Big(\frac{\nu_c}{2}+\frac{I_0(c)}{200\sinh c}\Big)x + a(x),$$

where $a(x)=e^{2\varepsilon}\log(e^{\varepsilon}x)\big(2.01\varepsilon\sqrt{x}+\frac12\log\log(2x^2)\big)$. *Moreover* $2\sqrt{c}\,e^{\varepsilon/2}\big(\nu_c/2+I_0(c)/(200\sinh c)\big)\le 0.80319<0.7946\cdot 1.02$ *for these* c *(the scan uses only integer* c*).*

Proof. Since ϕ is a probability density and ψ is non-decreasing,

$$\psi(x)-\psi_{c,\varepsilon}(x)=\int(\psi(x)-\psi_0(xe^{-\tau}))\phi(\tau)\,d\tau\le\int_0^{\varepsilon}\big(\psi(x)-\psi((xe^{-\tau})^-)\big)\phi(\tau)\,d\tau.$$

For $\tau<0$ the integrand is ≤ 0, because $\psi_0(y)\ge\psi(x)$ for $y>x$.

For $3\le Y\le X/2$ the Brun–Titchmarsh inequality of Montgomery and Vaughan, $\pi(X)-\pi(X-Y)\le 2Y/\log Y$ [20], together with the trivial count of prime powers as in [8, (3.9)], gives

$$\psi(X)-\psi(X-Y)\le\log X\Big(\frac{2Y}{\log Y}+0.6\frac{Y}{\sqrt{X}}+\log\log X^2\Big).$$

Put $Y = x(1-e^{-\tau}) \le x\tau$. For $\tau \ge \tau_0 := \varepsilon/(10\sqrt{c})$ we have $\log Y \ge \log x - \lambda_s$. For $\tau < \tau_0$ we bound the integrand by its value at τ_0, using monotonicity.

The mass of ϕ below τ_0 is at most $\tau_0 \max\phi \le \tau_0 e^{\varepsilon/2} c I_0(c)/(2\varepsilon\sinh c)$. Also $\int_0^\varepsilon \tau\phi \le e^{\varepsilon/2}\varepsilon\nu_c/2$, since $\lambda \ge 2$. Combining these gives the main term, with $\log x$ in the numerator.

The lower bound $\psi_{c,\varepsilon}(x) - \psi(x) \le \int_{-\varepsilon}^0 (\psi_0(xe^{|\tau|}) - \psi(x))\phi$ is treated in the same way. Here the intervals end at xe^{ε}, which produces the factor $\log(e^{\varepsilon}x) \le \log x + \lambda_s$. The prime-power and $\log\log$ terms give $a(x)$.

The final numerical inequality is certified in ball arithmetic by `certify/smoothing_const.py`, using `acb.integral` for ν_c. The maximum over c is attained at $c = 110$. □

Remark 3.2. [7, Prop. 4] (quoted as [8, Prop. 1]) states the main term as $2\varepsilon x|\nu_c(\alpha)|\log(e^{\varepsilon}x)/\log B$, with $\nu_c(t) = \int_{-\infty}^t \mu_c$ and $\mu_c(t) = -\int_{-\infty}^t \eta_{c,1}$ for $t < 0$. With $\eta_{c,1}$ normalised as above ($\int \eta_{c,1} = 2$), one has $|\nu_c(0)| = \nu_c \sim \sqrt{2/(\pi c)}$, whereas [8, Remark 1] quotes $|\nu_c(0)| \sim (2\pi c)^{-1/2}$, which corresponds to a density of mass 1. We have not been able to reconcile the two normalisations. Our own proof of the smoothing step (Lemma 3.1) does not depend on this. Its main constant, $0.80/\sqrt{c}$, is half of what $2|\nu_c(0)|$ would give with the stated normalisation.

3.2. **Explicit formula and truncation.**

Proposition 3.3 ([8, Prop. 2], [7, Props. 2,3])**.** *Let $x \ge 10$, $0 < \varepsilon \le 10^{-4}$ and $c \ge 3$. Then*

$$x - \psi_{c,\varepsilon}(x) = \frac{1}{\ell_{c,\varepsilon}(i/2)} \sum_{\rho} \ell_{c,\varepsilon}\Big(\frac{\rho - 1/2}{i}\Big)\frac{x^\rho}{\rho} + \Theta(2),$$

where $\Theta(2)$ denotes a quantity of absolute value at most 2, and

$$\sum_{|\operatorname{Im}\rho| > c/\varepsilon} \Big|\frac{\ell_{c,\varepsilon}((\rho-\frac12)/i)}{\ell_{c,\varepsilon}(i/2)}\frac{x^\rho}{\rho}\Big| \le 0.16\,\frac{x+1}{\sinh c}\, e^{0.71\sqrt{c\varepsilon}}\log(3c)\log\frac{c}{\varepsilon}.$$

Lemma 3.4. *Let $\rho = \beta + i\gamma$ with $0 < \beta < 1$ and $|\varepsilon\gamma| \le c$. Then*

$$\Big|\ell_{c,\varepsilon}\Big(\frac{\rho-1/2}{i}\Big)\Big| \le e^{\varepsilon(c+1)/6}\,\ell_{c,\varepsilon}(\gamma), \qquad \ell_{c,\varepsilon}(i/2) \ge 1.$$

Proof. Write $\ell_{c,\varepsilon}(z) = \frac{c}{\sinh c}\,G(c^2 - \varepsilon^2 z^2)$ with $G(w) = \sum_{n\ge0} w^n/(2n+1)!$. For $z = \gamma - i(\beta - \frac12)$ we have $|c^2 - \varepsilon^2 z^2| \le (c^2 - \varepsilon^2\gamma^2) + \varepsilon^2/4 + \varepsilon^2|\gamma| \le (c^2 - \varepsilon^2\gamma^2) + \varepsilon(c+1)$. Since the coefficients of G are positive, $|G(w)| \le G(|w|)$. The bound $\frac{d}{dw}\log G(w) \le 1/6$ for $w \ge 0$ then gives the first claim. The second holds because G is increasing. □

Lemma 3.5. *For $T \ge 10^4$, $\sum_{0<\gamma\le T} 1/\gamma \le \frac12\log^2 T$, the sum running over zeros with multiplicity.*

Proof. Write the sum as a Stieltjes integral $\int_{14}^T t^{-1}dN(t) = N(T)/T + \int_{14}^T N(t)t^{-2}dt$. Insert the explicit Riemann–von Mangoldt bound $|N(t) - \frac{t}{2\pi}\log\frac{t}{2\pi e} - \frac78| \le 0.1038\log t + 0.2573\log\log t + 9.3675$ for $t \ge e$ [14]. This gives $\sum_{0<\gamma\le T} 1/\gamma \le \frac{1}{4\pi}\log^2 T + \log T + 2$, which is below $\frac12\log^2 T$ for $T \ge 10^4$. At $T = H_0$ the bound is 96.5, against $\frac12\log^2 H_0 = 412.7$. □

3.3. **The bound for ψ on an interval.**

For $s \in [0.6, 1)$, $H = H_0 = 3\cdot10^{12}$ and $T_1 = c/\varepsilon$, put

$$S(s) := \sum_{\substack{H<\gamma\le T_1\\ \beta>s}} \frac{\ell_{c,\varepsilon}(\gamma)}{\gamma}\, e^{\varepsilon(c+1)/6}.$$

Proposition 3.6. *Let $43 \le L_1 < L_2$, $c \in [30, 110]$ and $\varepsilon \le 10^{-4}$, with $\lambda_s < L_1 - 1$. Let $T_1 = c/\varepsilon$. Assume every zero $\beta + i\gamma$ with $H_0 < \gamma \le T_1$ has $\beta \le s_{\max} := 1 - \lambda^*(\log T_1)/\log T_1$, where λ^* is as in §4 (this holds by Theorem 4.10). Then for every x with $\log x \in [L_1, L_2]$,*

$$\frac{|\psi(x) - x|}{x} \le \frac{L_2 + \lambda_s}{L_1 - \lambda_s} \cdot \frac{0.7946 \cdot 1.02\,\varepsilon}{\sqrt{c}} + \frac{0.32\, e^{0.71\sqrt{c\varepsilon}} \log(3c) \log T_1}{\sinh c} + e^{-L_1/2} \log^2 H + 2\Big(L_2 \int_{0.6}^{s_{\max}} e^{(s-1)L_1} S(s)\, ds + e^{-0.4L_1}\big(S(0.6) + \log^2 T_1\big)\Big) + \frac{a(x) + 2}{x}.$$

Proof. Combine Lemma 3.1 with Proposition 3.3. The zeros with $|\gamma| > T_1$ contribute the second term, since $0.16(x+1) \le 0.32x$.

Zeros with $|\gamma| \le H$ lie on the critical line [24]. By Lemmas 3.4 and 3.5 they contribute at most $x^{1/2} \cdot 2 \cdot \frac{1}{2} \log^2 H$.

Zeros with $H < |\gamma| \le T_1$ come in pairs $\rho, \bar\rho$, which gives the factor 2. For those with $\beta > 0.6$, use the identity

$$\sum_{\beta > 0.6} x^{\beta - 1} f(\gamma) = x^{-0.4} S(0.6) + \log x \int_{0.6}^{1} x^{s-1} S(s)\, ds,$$

together with $x^{s-1} \le e^{(s-1)L_1}$ and $\log x \le L_2$; also $S(s) = 0$ for $s \ge s_{\max}$.

All remaining zeros in this height range have $\beta \le 0.6$. These include the reflections $1 - \bar\rho$ of zeros off the line, and the zeros on the line. They contribute at most $x^{-0.4} e^{\varepsilon(c+1)/6} \sum_{H < \gamma \le T_1} 1/\gamma \le x^{-0.4} \log^2 T_1$; here Lemma 3.5 gives $\frac{1}{2} \log^2 T_1$, which leaves ample room for the factor $e^{\varepsilon(c+1)/6} \le 1 + 10^{-9}$. □

Lemma 3.7 (Stieltjes bound)**.** *Let $f(t) = \ell_{c,\varepsilon}(t)/t$, which is decreasing on $(0, T_1]$. Let $H = t_0 < t_1 < \cdots < t_n = T_1$, and let $N^+(s,t)$ be any upper bound for $N(s,t) := \#\{\rho : \beta > s,\ 0 < \gamma \le t\}$. Suppose $N^+(s, t_{i+1})$ is replaced by 0 whenever $s \ge 1 - \lambda^*(\log t_{i+1})/\log t_{i+1}$. Then*

$$S(s) e^{-\varepsilon(c+1)/6} \le \sum_{i < n} N^+(s, t_{i+1})(f(t_i) - f(t_{i+1})) + N^+(s, T_1) f(T_1).$$

Proof. This is summation by parts. Zeros with $\gamma \le H$ do not satisfy $\beta > s \ge 0.6$, so they do not contribute. The zero-free region is used at the top of each block; this is valid because $\lambda^*(l)/l$ is non-increasing in l. □

In the computation, $\int_{0.6}^{s_{\max}} e^{(s-1)L_1} S(s)\, ds$ is bounded by $\sum_j S(s_j) \int_{s_j}^{s_{j+1}} e^{(s-1)L_1} ds$, which is valid because S is non-increasing. The grid consists of 61 points up to $s_{\max} - 0.02$, followed by steps $1/(40L_1)$. The function N^+ is the minimum of the density bounds of §6.

4. A height-dependent zero-free region

4.1. **The inequality of Bellotti, Trudgian and Yang.** We recall the set-up of [5] and keep its notation.

- $\theta = 1.1338$, and w is Heath-Brown's function [5, Def. 1], supported on $[0, D]$ with $D = 2\theta \cot\theta$. Its Laplace transform is $W(z) = \int_0^D e^{-zu} w(u)\, du$, and $w_0 = w(0) = 5.672787598\ldots$.
- κ_m $(0 \le m \le 6)$ are the coefficients in [5, (18)], with $\kappa = \sum \kappa_m = 433/859$.
- $P(x) = \sum_{k \le 16} a_k \cos kx \ge 0$ is the trigonometric polynomial of [5, §4], with $a_0 = 1$, $a_1 = 865534/497079$ and $a = \sum_{k \ge 1} a_k = 2919857/828465$.
- $K = 16$ and $T_0 = 10^{10}$.

For $\eta > 0$ and $\frac{1}{2} < \sigma < 1$, put

$$f(u) = \eta w(\eta u) \sum_m \kappa_m e^{-(2\sigma-1)mu}, \qquad F(z) = \sum_m \kappa_m W\big((z + (2\sigma-1)m)/\eta\big).$$

In particular $f(0) = \kappa\eta w_0$.

Let $\rho_0 = \beta_0 + it$ be a zero with $t \geq H_0$, and put $\eta = 1 - \beta_0$ and $\lambda_0 = \eta \log t$. Suppose ζ has no zeros with $\beta \geq 1 - A/\log\gamma$ for $H_0 \leq \gamma \leq Kt + T_0$, and put $\sigma = 1 - A/\log(Kt + T_0)$. Assume $\eta \leq \eta_0 \leq 0.0145$ and $\sigma \geq \sigma_0 \geq 0.9930$. The argument of [5, Lemma 15] then gives

$$c_{10}\,\eta w_0 \leq \eta w_0\Big(\frac{a\kappa}{2}\log t - G\Big) + a_0 F(\sigma-1) - a_1\big(F(\sigma-1+\eta) + F(\sigma-\eta)\big) - Z + 10^{-10}\eta + 2\cdot 10^{-7} + E_2\lambda_0\eta^2 + E_3\eta^3. \tag{4.1}$$

Here $Z \geq 0$ is the part of the zero sum that is discarded. The constants c_{10}, G, E_2, E_3 depend on (η_0, σ_0) and are given in §4.2. A zero with these parameters is impossible as soon as the following margin is positive:

$$\mathcal{M}_1 := a_1\big(F(\sigma-1+\eta) + F(\sigma-\eta)\big) - \Big[\eta w_0\big(\tfrac{a\kappa}{2}\log t - G - c_{10}\big) + a_0 F(\sigma-1) + 10^{-10}\eta + 2\cdot 10^{-7} + E_2\lambda_0\eta^2 + E_3\eta^3\Big]. \tag{4.2}$$

Remark 4.1. In [5] the Lemma-10 term enters the final inequality as $-0.1186\,w(0)\eta$, i.e. with $f(0) = \eta w(0)$. By their Definition 2, $f(0) = \kappa\eta w(0)$ with $\kappa = 0.504$. With the correct value, the Lemma-10 gain $0.23545\,\kappa f(0)$ equals $0.0598\,\eta w_0$, not $0.1186\,\eta w_0$. Evaluating their inequality exactly at the critical height $t = e^{76.47}$ with this correction still gives $\eta\log t \geq 0.2054 > 1/4.896$. So their Theorem 1 is unaffected. (We also note the misprint $F(2\sigma-1+\beta_0)$ in their (25), which should read $F(\sigma-1+\beta_0)$, as in (47).)

4.2. **The constants for general (η_0, σ_0).** The constants of [5] are computed for $\eta \leq 0.0071093$ and $\sigma \geq 0.9935164$. We need η up to about 0.0125 and σ down to 0.9930, so we re-derive each (η_0, σ_0)-dependent ingredient.

Lemma 4.2 (W_0 bound; replaces [5, Lemma 4])**.** *Put $W_0(z) = W(z) - w_0/z$. For $\operatorname{Re} z \geq \nu$ and $|z| \geq r > 0$,*

$$|W_0(z)| \leq C_w(\nu, r)\,|z|^{-3}, \qquad C_w(\nu, r) = |w''(0)| + \frac{|w'''(0)| + e^{-\nu D}|w'''(D)| + \int_0^D |w''''(u)|e^{-\nu u}\,du}{r}.$$

Here $|w''(0)| = 50.06970149\ldots$ and $|w'''(0)| = |w'''(D)| = 117.26477652\ldots$.

Proof. We have $w(D) = w'(D) = w''(D) = 0$ and $w'(0) = 0$. Integrating by parts four times gives $W(z) = \frac{w_0}{z} + \frac{w''(0)}{z^3} + \frac{w'''(0) - e^{-zD}w'''(D)}{z^4} + z^{-4}\int_0^D e^{-zu}w''''(u)\,du$. The derivative values are exact enclosures from Taylor series in `arb`. The integral is certified on an integer grid of $\nu \in [30, 200]$, using that it is decreasing in ν. Sign changes of w'''' are isolated, and each piece is integrated with `acb.integral` (`rep/consts_cert.py`). □

For comparison, $C_w(69.4, 69.4) = 51.80$, while [5] use $C < 52$ at this point.

Lemma 4.3 (non-negativity; replaces [5, Lemma 5, (21)])**.** *Let $\sigma \in [0.993, 1)$ and $\eta \leq \eta_0 \leq 0.0145$. Then $\operatorname{Re} F(z) + \operatorname{Re} F(2\sigma - 1 - \bar{z}) \geq 0$ for $0 \leq \operatorname{Re} z \leq 2\sigma - 1$.*

Proof. As in [5], it suffices to treat $\operatorname{Re} z = 0$. For fixed σ every δ_m equals $(2\sigma-1)m$ exactly. Rescaling $y \mapsto y/(2\sigma-1)$ reduces the condition to

$$B(y) := \sum_{m=1}^{7} b_m \frac{m - \operatorname{sgn}(b_m)\varepsilon_0/m}{m^2 + y^2} > 0 \quad (y \in \mathbb{R}), \qquad \varepsilon_0 = \frac{C_w\eta_0^2}{w_0(2\sigma-1)^2}.$$

Here $(1+x)\sum \kappa_m x^m = \sum b_m x^m$. This reduction is independent of σ. For $\eta_0 \le 0.0145$ and $\sigma \ge 0.993$ we have $\varepsilon_0 \le 1/500$ (with $C_w \le 51.9$). For $\varepsilon_0 = 1/500$, B is a rational function whose numerator has no real roots and whose value at 0 is positive; this is checked by exact root isolation (`rep/lemma5_cert.py`). □

Remark 4.4. [5] prove (21) uniformly in σ, replacing δ_m by $c_0 m$ with $c_0 = 151/153 \le 2\sigma_0 - 1$. This requires $\sigma \ge 0.993464$. For $c_0 = 0.9864$, which corresponds to the smaller values of σ needed below, their test function B takes negative values, already at $\varepsilon_0 = 1/2000$. The per-σ version above has no such restriction.

Lemma 4.5 (error terms; replaces [5, Lemmas 8, 9 and (48)]). *Let $x_0 = \sigma_0 - \frac12 \ge 0.493$, $C_1 = C_w(x_0/\eta_0, x_0/\eta_0)$ and $C_2 = C_w(\sigma_0/\eta_0, \sigma_0/\eta_0)$. Let $\Sigma_\kappa = \sum|\kappa_m|$. Then $\sum_k a_k|E_k| \le E_2\lambda_0\eta^2 + E_3\eta^3$, where*

$$E_2 = a\,\Sigma_\kappa \frac{C_1}{\pi x_0^2}, \qquad E_3 = a_0\Sigma_\kappa\Big(\frac{C_1}{2\pi}\cdot 23.6497 + \frac{C_2}{x_0^3}\Big) + a\,\Sigma_\kappa\frac{C_2}{x_0^3}.$$

Proof. Follow [5, Lemma 8], with two changes. First, in their (33) the inequality $\eta\log t \le A_0$ is replaced by $\eta\log t = \lambda_0$. Second, the integral (32) is evaluated with the true digamma function: $\int_{\mathbb R} |\operatorname{Re}\psi(\frac14 + \frac{iy}{2})|(x_0^2+y^2)^{-3/2}dy \le 23.6497$ at $x_0 = 0.4930$, certified (`rep/consts_cert.py`). This integral is decreasing in x_0. □

Remark 4.6. The majorant $U(y)$ of [5, (31)] integrates to 28.3 against $(x_0^2+y^2)^{-3/2}$, not to less than 24 as stated in (32). The true digamma gives 23.59, so (32) is correct, but its proof as written does not establish it. At $(\eta_0,\sigma_0) = (0.0071093, 0.9935164)$, Lemma 4.5 gives $185.4\eta^2 + 7972\eta^3$, against $187\eta^2 + 8013\eta^3$ in [5, (48)].

Lemma 4.7 ([5, Lemmas 3, 10, 11], re-derived). (1) *$f \ge 0$ on $[0,\infty)$.*

(2) *$f(u) \ge \kappa f(0)$ for $0 \le u \le u_0(\eta_0,\sigma_0)$; for example $u_0 = 59.15$, 49.46 and 33.61 at $\eta_0 = 0.0071$, 0.0085 and 0.0125.*

(3) *$\sum_n \Lambda(n)n^{-\sigma}f(\log n)P(t\log n) \ge c_{10}\,\eta w_0$, where $c_{10} = \kappa^2\sum_{p:\ 15\log p\le u_0} m_p \log p$.*

Here m_p are the certified lower bounds for $\min_x G_p(1,x)$ in [5, Table 1], except that $m_5 = 0.0073564$ (Remark 4.8).

Proof. (1) $w \ge 0$ on $[0,D]$; this is certified by ball bisection on $[0, D-0.02]$ and by a Taylor bound near D, where $w'''(D) < 0$. The polynomial $\sum \kappa_m x^m$ has no root in $[0,1]$, by exact root isolation (`rep/f_nonneg_cert.py`).

(2) We have $g(u) := w(\eta_0 u)\sum \kappa_m e^{-d_m u} - \kappa w_0 \ge 0$, with d_m as in [5, Lemma 11]. This is certified by ball bisection, using $g(0) = 0$ and $g'(0) = -w_0\sum \kappa_m d_m \ge 1.19$ near $u = 0$ (`rep/lemma11_cert.py`).

(3) As in [5, Lemma 10], the terms with $n > e^{u_0}$ are dropped, which is valid by (1) and $P \ge 0$. So are all primes with $15\log p > u_0$, and the powers p^m with $m > 15$. The bound $G_p(\sigma,x) \ge G_p(1,x)$ needs no computation: $G_p(\sigma,x) = \sum_{m\le 15} p^{-m\sigma}P(mx)$, so $\partial_\sigma G_p = -\log p\sum_m m\,p^{-m\sigma}P(mx) \le 0$. The values m_p are re-certified by ball bisection with a mean-value enclosure (`rep/mp_cert.py`). □

Remark 4.8. For $p = 5$, [5, Table 1] lists $m_5 = 0.00735709$. However $G_5(1,x)$ attains $0.0073564919\ldots$ at $x = 1.6495242\ldots$, as confirmed independently in 30-digit arithmetic. All other entries of the table are confirmed. The effect on their constant 0.23545 is -10^{-6}.

Lemma 4.9 ([5, Lemma 12]). $\sum_k a_k \sum_m \kappa_m\big(-\frac12\log\pi + \frac12\operatorname{Re}\frac{\Gamma'}{\Gamma}(\frac{s_k+\delta_m}{2}+1)\big) \le \frac{a\kappa}{2}\log t - G(\sigma_0)$, *where $G(\sigma_0)$ is computed by their formula in ball arithmetic. We have $G(0.9930) = 1.56839$ and $G(0.9935164) = 1.56888$; [5] give 1.568.*

Lemma 13 of [5] ($10^{-10}\eta$) and the tail estimates in their Lemma 6 hold unchanged for $\eta \le 0.015$ and $\log t \le 45$. Each tail term in their Lemma 6 is at most $3 \cdot 10^{-9}$, so the total stays below 10^{-7}; we use $2 \cdot 10^{-7}$.

4.3. **Exact evaluation of W.** Since w is a combination of $u\cos(ku)$, $\cos(ku)$, $\sin(\phi - ku)$, u and 1 on $[0, D]$ (with $k = \tan\theta$), $W(z)$ has a closed form. In terms of $J_0(s) = (1 - e^{-sD})/s$ and $J_1(s) = (1 - e^{-sD}(1 + sD))/s^2$ it reads

$$W(z) = \sec^2\theta\Big[\sec^2\theta \cdot \tfrac{1}{2}\sum_{\pm}\big(\theta\cot\theta\, J_0 - \tfrac{1}{2}J_1\big)(z \mp ik) + (2\theta\cot\theta - 2)J_0(z) - J_1(z) + \frac{\Sigma_{2\theta}(z)}{\sin 2\theta} - \frac{2\Sigma_\theta(z)}{\sin\theta}\Big],$$

where $\Sigma_\phi(z) = (e^{i\phi}J_0(z + ik) - e^{-i\phi}J_0(z - ik))/(2i)$. It agrees with quadrature to 10^{-15} (`rep/Wexact.py`). In certification we evaluate it in `acb`, with the second-order enclosure

$$W(z) \in W(z_c) + W'(z_c)(z - z_c) + B\big(0, \tfrac{1}{2}r^2 \sup|W''|\big), \qquad r = |z - z_c|.$$

Here W' has the same closed form with $J_0 \to -J_1$ and $J_1 \to -J_2$. Integrating $u^2 w$ by parts three times gives $\sup|W''| \le \min\big(\int u^2 w e^{-\nu u},\ (2w_0 + \int |(u^2 w)'''|e^{-\nu u})/|z|^3\big)$. Both integrals are bounded by sums of ball suprema over 256 subintervals.

4.4. **The zero-free region.**

Theorem 4.10. *Let $\lambda^*(l)$ be the function given by the table below on the grid $l \in \{28, 28.25, \dots, 43\}$, extended by $\lambda^*(l) := \lambda^*(\lceil l \rceil_{0.25})$ for $l \le 43$ and by $\lambda^*(l) := 1/4.896$ for $l > 43$. Then every zero $\beta + i\gamma$ of ζ with $\gamma > H_0$ satisfies $(1 - \beta)\log\gamma \ge \lambda^*(\log\gamma)$; for $3 \le \gamma \le H_0$ all zeros have $\beta = 1/2$ [24].*

l	28	29	30	31	32	33
$\lambda^*(l)$	0.21424	0.21384	0.21345	0.21308	0.21272	0.21239
l	34	35	36	37	38	40
$\lambda^*(l)$	0.21206	0.21175	0.21145	0.21117	0.21084	0.21033

The full table is `rep/lamstar2c.json`.

Proof. For $\gamma \le H_0$ the statement follows from [24]. For $l > 43$ it is [5, Thm. 1]. For the remaining range we run a four-step chain, starting from $\lambda_0 \equiv 1/4.896$.

At step j the hypothesis is the conclusion of step $j - 1$. Since $\lambda_{j-1}(l)/l$ is non-increasing, this hypothesis gives the zero-free input of Lemma 6 of [5] with $A = \lambda_{j-1}(\log(Kt + T_0))$. In every height cell $(L - 0.25, L]$ we certify $\mathcal{M}_1 > 0$ in ball arithmetic. This is done for all $\log t$ in the cell and all $\lambda_0 = \eta\log t \in [\lambda_{j-1}(L), \lambda_j(L)]$, where $\lambda_j(L)$ is the float threshold minus $2 \cdot 10^{-5}$, by adaptive bisection of the box. On each box we take σ as a ball, with A at the top of the box, which is valid because a smaller A is a weaker hypothesis. The constants are taken at $\eta_0 = \lceil \lambda_j/l_{\min} \rceil_{10^{-4}} + 10^{-4}$ and at σ_0 rounded down. Then $\lambda_j(L) \ge \lambda_{j-1}(L)$, and a zero with $\lambda_{j-1}(L) \le (1 - \beta)\log\gamma < \lambda_j(L)$ would give $\mathcal{M}_1 > 0$, contradicting (4.1).

We use the table produced by the fourth step; it differs from the third by at most 10^{-5} in every entry. Program: `rep/lamcert.py`. □

At $\log t = 29$ the theorem amounts to the zero-free region $\sigma > 1 - 1/(4.676\log t)$.

5. Repulsion of zeros near the 1-line

In [5, Lemma 6] the zero sum $\mathrm{Re}\sum_\rho F(s_1 - \rho)$ is bounded below by the pair $\rho_0, 1 - \bar\rho_0$. Every other zero ρ is paired with its reflection $1 - \bar\rho$, and each such pair contributes ≥ 0 by (21) (our Lemma 4.3). Instead of discarding one more pair, we keep it.

Lemma 5.1. *In the situation of (4.1), suppose $\rho_1 = \beta_1 + i(t+d)$ is a second zero with $\beta_1 > 1/2$. Put $\eta_1 = 1 - \beta_1$, and assume λ^*-admissibility, i.e. $\sigma - \beta_1 > 0$. Then (4.1) holds with*

$$Z \geq a_1 \operatorname{Re}\big(F(\sigma - 1 + \eta_1 - id) + F(\sigma - \eta_1 - id)\big).$$

Proof. The pair $\{\rho_1, 1 - \bar{\rho}_1\}$ contributes exactly $\operatorname{Re} F(s_1 - \rho_1) + \operatorname{Re} F(s_1 - (1 - \bar{\rho}_1))$ to the zero sum, and this equals the displayed quantity. All other pairs are ≥ 0 by Lemma 4.3, since $0 \leq \operatorname{Re}(s_1 - \rho) \leq 2\sigma - 1$ for every zero below height $Kt + T_0$. The term s_1 enters with the coefficient a_1. The case $d < 0$ is identical, because $F(\bar{z}) = \overline{F(z)}$, so only $|d|$ matters. □

A pair of zeros at heights $t, t + d$ is therefore impossible as soon as

$$\mathcal{M}_2 := \mathcal{M}_1(\eta_0) + a_1 \operatorname{Re}\big(F(\sigma - 1 + \eta_1 - id) + F(\sigma - \eta_1 - id)\big) > 0$$

for at least one of the two choices of primary zero. Here F is formed with the primary η. The reflected term behaves like $\operatorname{Re} F(\sigma - \eta_1 - id) \approx \eta w_0(\sigma - \eta_1)/((\sigma - \eta_1)^2 + d^2)$, which is of size about 0.04 for $d \lesssim 0.05$ and $\eta, \eta_1 \approx 0.0075$. This is what produces the repulsion.

Theorem 5.2. *For integers $29 \leq l_k \leq 37$ and k with $\lambda^*(l_k) < k \cdot 10^{-4} \cdot l_k \leq 0.0125(l_k - 1)$, define $d^*(l_k, k)$ by the table `rep/dstar2c_all.json`. For each l_k the value is largest for the smallest admissible k (for example 0.142 at $l_k = 29$), and for the large values of k it equals 0.058 ($l_k = 29$), 0.056 ($l_k = 30$) and 0.045 ($l_k = 37$). Suppose two zeros $\beta_j + i\gamma_j$ ($j = 0, 1$) satisfy $H_0 < \gamma_j \leq e^{l_k}$ and $1 - \beta_j \leq k \cdot 10^{-4}$. Then $|\gamma_0 - \gamma_1| \geq d^*(l_k, k)$.*

Proof. For a cell (l_k, k) put $e_{\min} = \lambda^*(l_k)/l_k$ and $e_{\max} = k \cdot 10^{-4}$; the variables η_0, η_1 below denote $1 - \beta$ of the two zeros. By Theorem 4.10, both zeros have $1 - \beta_j \geq e_{\min}$, since $\lambda^*(l)/l$ is non-increasing.

Fix a sub-cell $(l - 1, l]$ of heights, $l \leq l_k$. We certify $\mathcal{M}_2 > 0$ for all $(\log t, \eta_0, \eta_1, |d|) \in [l - 1, l + 10^{-12}] \times [e_{\min}, e_{\max}]^2 \times [0, d^*]$, in ball arithmetic by adaptive bisection of the box. The width 10^{-12} covers the height $t + d$ of the second zero. On each box we try both orderings of the pair. As before, σ is taken as a ball, with A at the top of the box, and the constants at (η_0, σ_0) rounded outward. In this way all k on a coarse grid are certified at 97% of the float value of d^* (`rep/rep_cert.py`, `rep/repcert_run.py`). Intermediate k inherit the value of the next certified k above, which is valid because enlarging $e_{\max}$ only enlarges the box.

Finally, a zero counted at height $\leq e^{l_k}$ may lie in any lower height cell. So the table records $\min_{l \leq l_k} d^*(l, k)$, over the cells l in which $1 - \beta \leq k \cdot 10^{-4}$ is compatible with Theorem 4.10 (`rep/min_fix.py`). This minimum changes only the six cells $(l_k, 121)$ with $l_k \geq 32$, which become 0. □

The first version of these certificates used a second-order enclosure for W'' that clamped ν from below at -2. An independent audit found this. After the fix, every one of the 131 certified cells, and the chain of Theorem 4.10, was certified again with identical results.

6. Zero-density estimates

Let $N(\sigma, T) = \#\{\rho = \beta + i\gamma : \beta > \sigma,\ 0 < \gamma \leq T\}$, counted with multiplicity. In Proposition 3.6, $N(s, t)$ is replaced by $N^+(s, t)$, the minimum of the following bounds, each valid in its stated range.

- (N1) $N(T) \leq \frac{T}{2\pi} \log \frac{T}{2\pi e} + \log T + 1$ for $T \geq H_0$; this follows from [14].
- (N2) For $\sigma \geq \sigma_0 \in \{0.60, 0.65, \ldots, 0.99\}$ and $T \geq 3.061 \cdot 10^{10}$: Kadiri–Lumley–Ng [19, Table 1], in the form $N(\sigma, T) \leq A(\log T)^{5 - 2\sigma_0} T^{\frac{8}{3}(1 - \sigma_0)} + B \log^2 T$. This is valid because $\sigma \geq \sigma_0$, and the rows with $k < 1$ are used as $k = 1$.
- (N3) For $\sigma \geq 0.9927$ and $T > H_0$: Bellotti's bounds $N(\sigma, T) \leq C T^{B(1 - \sigma)}$ [4, Thm. 1.1, Table 2], uniformly and for the ranges $T \in (e^{l-1}, e^l]$ $(29 \leq l \leq 41)$.

(N4) For $1-\sigma \le 0.03$ and $28 < \log T \le 40$: Theorem 6.1 below. It is this bound that matters in the binding range.

6.1. **The log-free estimate.** We follow Bellotti [4, §§3–4]. In this section $L = \log T$ (not $\log x$), and $U = T^u$, $V = T^v$ ($v = \tau u$), $W = T^w$ and $X = T^x$.

- The Barban–Vehov weights are ψ_d, supported on $d \le V$, with $\Psi(n) = \sum_{d|n} \psi_d$, so that $\Psi(1) = 1$ and $\Psi(n) = 0$ for $1 < n \le U$.
- The Selberg-type weights are θ_d, supported on $d \le W$, with $\Theta(n) = \sum_{d|n} \theta_d$.

Let $\alpha = 1 - \delta$.

- The plane is cut into rectangles of height $h = \kappa_0 \delta$.
- In each rectangle we choose one representative zero with $\beta \ge \alpha$, if there is one; let J be the number of representatives.
- Let $\tilde{n}$ be an upper bound for the number of zeros with $\beta \ge \alpha$ in one rectangle.

Theorem 6.1. *Let $\delta \in ((k-1)10^{-4}, k\,10^{-4}]$ with $k \le 300$, and $\log T \le l$ with $l \in \{29, 29.5, 30, 30.5, 31, 32, \dots, 40\}$. Let (u, τ, w, x) be admissible for the cell (k, l) in the sense of Lemma 6.2. Then*

$$N(1-\delta, T) \le N(1-\delta_k, e^l) \le \tilde{n}\,\frac{d_4 d_5}{w} \cdot \frac{\mathcal{C}}{c_1^2 - c_5} \cdot \frac{e^{2x\delta_k l}}{\delta_k l},$$

where $\delta_k = k\,10^{-4}$ and the constants $\tilde{n}, d_4, d_5, \mathcal{C}, c_1, c_5$ are those of Lemma 6.2, evaluated at δ_k and $L = l$. (The first inequality is monotonicity of N; no rescaling within a cell is used.)

The parameters are chosen separately for each cell by minimising the bound numerically. Any admissible choice gives a valid bound, so the optimisation needs no certification. The bound is applied at the corner $(1-\delta_k, e^l)$ of the cell, which dominates the whole cell by monotonicity. The resulting tables are `code/lemmaD_tableGH0.01_*.pkl`.

Lemma 6.2. *Write $\alpha_k = 1 - \delta_k$ and $\nu = \lambda^*(l)/l$. Call (u, τ, w, x) admissible if*

$$0.02 < w < u, \quad \tau > 1.05, \quad xL > uL - 2\log L, \quad uL \ge 40, \quad xL > \tau u L,$$

$$\delta_k > \nu, \quad c_1 > 0, \quad c_1^2 - c_5 \ge 0.2.$$

For admissible parameters the bound of Theorem 6.1 holds with the following constants.

(i) $c_1 = e^{-L^2/U} - B_1 - 3.8\,e^{-L(L-2u)}\dfrac{L^2+1}{w(v-u)L^6}$, *where* $I_\Gamma = 5.922$ *and*

$$B_1 = \frac{10.45 \cdot 0.618 \cdot L \cdot I_\Gamma}{2\pi}\, T^{(v+w)/2 + 1/6 + x(1/2-\alpha_k)}.$$

(ii) $d_5 = 1.0061\big(1 + (1.333 + 4e^{-wL/2})/(wL)\big)$ [4, Lemma 4.3].

(iii) d_4 *is any upper bound for* $\sum_n \Psi(n)^2 n^{-1}(e^{-n/X} - e^{-nL^2/U})$; *we take the bound of Theorem 6.5.*

(iv) $c_5 = \dfrac{d_4 d_5}{2\pi}\,\dfrac{e^{wL}}{wL}\Big(e^{Lx(3/2-2\alpha_k)} + e^{(Lu-2\log L)(3/2-2\alpha_k)}\Big)\,\mathcal{J}$, *where* $\mathcal{J} = 4.72 \cdot 0.618\,e^{L/6}L$.

(v) $\mathcal{C} = \min_{\kappa_0}\Big[1.0067\big(1 + 2\delta_k\,\Lambda_{\mathrm{pair}}(\kappa_0\delta_k)\big)\Big] + 2\sqrt{2\pi}\cdot 2.4 \cdot I'(\delta_k, l)$, *with* Λ_{pair} *from Lemma 6.3 and I' from Lemma 6.4.*

(vi) $\tilde{n} = \max\big(1, \min_\eta \lfloor M(\eta, l)/\mathrm{cost}(h/2)\rfloor\big)$ *with $h = \kappa_0\delta_k$, in the notation of Lemma 6.3.*

Proof. We indicate only where the argument departs from [4]. The overall structure is [4, (4.22)–(4.23)]: $J^2 c_1^2 \le J(c_2 + c_3 + c_4)X^{2-2\alpha}/(\delta \log T) + J^2 c_5$, followed by $N(\alpha, T) \le \tilde{n} J$. Each change below replaces one of her constants by a smaller valid one.

(i) The term B_1. This replaces Bellotti's $44.66\,d_2(v+w)^4T^{(v+w)/2+x(1/2-\alpha_0)+27/164}\log^4 T$ in [4, (3.17)]. Write $a=ga'$ and $b=gb'$ with $(a',b')=1$, and use $|\psi_a|,|\theta_b|\le 1$ and $\sum_{n\le N} n^{-1/2}\le 2\sqrt{N}$. This gives $|M(\frac12+it)|\le \sum_g g^{-1/2}\cdot 2(V/g)^{1/2}\cdot 2(W/g)^{1/2} < 4\zeta(3/2)\sqrt{VW}=10.45\sqrt{VW}$, a bound without logarithms. We also use $|\zeta(\frac12+iu)|\le 0.618|u|^{1/6}\log|u|$ for $3\le|u|\le e^{105}$ [16], and ≤ 1.461 for $|u|<3$ [15], together with $(\gamma+|t|)^{1/6}\log(\gamma+|t|)\le T^{1/6}\log T\,(1+|t|)$. Finally $I_\Gamma=\sup_{\sigma\in[-\frac12,\frac12-\alpha]}\int|\Gamma(\sigma+it)|(1+|t|)\,dt\le 5.922$ for $\delta\le 0.03$. The last term of c_1 is Bellotti's bound for the tail $\sum_{n>U}\Psi\Theta n^{-\rho}e^{-nL^2/U}$ [4, (3.19)ff.].

(iv) The term $\mathcal{J}$. This replaces $J_1+J_2+J_3+J_4\le 0.241+5.921+286.077\,T^{27/164}+0.241$ [4, (4.17)–(4.19)]. We have $c:=\frac32-\beta_r-\beta_s\in[-\frac12,-0.44]$ for $\delta\le 0.03$, and $|\zeta(\frac12+iu)|\le \max(1.461, 0.618|u|^{1/6}\log|u|)$ for all $|u|\ge 3$. For $|u|>e^{105}$ this holds because Patel and Yang's $66.7|u|^{27/164}$ [22] is smaller there. Moreover $|u|\le T+|t|$. Hence the whole integral [4, (4.16)] is at most

$$0.618\,T^{1/6}\log T\int_{\mathbb{R}}|\Gamma(c+it)|\Big(1+\frac{|t|}{3}\Big)^{1/6}\Big(1+\frac{\log(1+|t|/3)}{\log 3}\Big)dt\le 4.72\cdot 0.618\,T^{1/6}\log T.$$

The integral equals 4.7136 at $c=-0.44$ and increases with c.

(iii) The diagonal. d_4 is the Barban–Vehov mean square of [4, Lemma 4.2], which we bound by Theorem 6.5. Bellotti's own bound [4, Lemma 3.3, (3.6)] gives $d_4\approx 85$–105 at her parameters (her Table 5); Theorem 6.5 gives about 1.5 at ours. Because Theorem 6.5 bounds the whole sum, Bellotti's condition $u+v<x$ is not needed.

(v) The off-diagonal terms c_2+c_3. Bellotti assumes [4, (4.11)] that representatives of distinct rectangles are at least δ apart. This need not hold: adjacent rectangles can have representatives arbitrarily close. Lemma 6.3 makes no such assumption, and it also uses the repulsion of Theorem 5.2. The far term c_4 is replaced using Lemma 6.4.

(vi) The local count. $\tilde{n}$ replaces $b_1\delta\log T+b_2$ [4, Lemma 2.4]. □

6.2. **The pair sum.** For $s_0=1+\eta+i\gamma_r$, the Hadamard product and Stirling's formula give

$$\sum_\rho \mathrm{Re}\,\frac{1}{s_0-\rho}\le M(\eta,L):=\min\Big\{\tfrac12\log\tfrac{T}{2\pi}+\mathcal{Z}(\eta),\ \min_{c\in[0.5,2]}\Big(\tfrac12\log\tfrac{T}{2\pi}+\tfrac{c^2+1/2}{2c}\mathcal{Z}(\eta)+\tfrac{\log(T/\pi)}{8c}\Big)\Big\},$$

where $\mathcal{Z}(\eta)=-\zeta'(1+\eta)/\zeta(1+\eta)$. The second expression comes from positivity of $(c+\cos\phi)^2$. Every term in the sum is positive. A zero with $\beta\ge 1-\delta$ at vertical distance d from s_0 contributes at least $\mathrm{cost}(d):=\min_{a\in\{\eta,\eta+\delta\}}a/(a^2+d^2)$, which is valid because $a\mapsto a/(a^2+d^2)$ is unimodal.

Lemma 6.3. *Fix a representative r and $h=\kappa_0\delta$. On each side of r, the representative of the j-th rectangle away lies at distance $d\in((j-1)h,(j+1)h)$. Its contribution to the Halász–Montgomery pair sum, relative to the diagonal, is at most $b_j=g\big(\max(2\delta,(j-1)h,d^*)\big)/\max\big(2\delta,(j-1)h,d^*\big)$, where $g(d)=\min\big(1,|\Gamma(1+2\delta+id)|/\Gamma(1+2\delta)\big)$ and $d^*=d^*(\lceil l\rceil,k)$ is the repulsion distance of Theorem 5.2. Its cost is at least $\mathrm{cost}((j+1)h)$. Rectangles with $(j+1)h\le d^*$ contain no representative. Then*

$$\sum_{s\ne r,\ |\gamma_s-\gamma_r|\le 1}\frac{|\Gamma(2-\rho_r-\bar\rho_s)|X^{2-\beta_r-\beta_s}}{1.0067\,X^{2\delta}}\le 2\delta\,\Lambda_{\mathrm{pair}}(h),$$

where $\Lambda_{\mathrm{pair}}(h):=\min_\eta\max\big\{\sum_j b_jx_j:0\le x_j\le 1$ (two per j), $\sum_j\mathrm{cost}((j+1)h)x_j\le M(\eta,L)-\mathrm{cost}(0)\big\}$.

Proof. We have $|\Gamma(z)|=|\Gamma(1+z)|/|z|$. For $0\le\mathrm{Re}\,z=s\le 2\delta$, $|\Gamma(1+s+id)|\le\Gamma(1+s)|\Gamma(1+2\delta+id)|/\Gamma(1+2\delta)$, since the product $\prod_n(1+d^2/(1+s+n)^2)^{-1}$ is increasing in s. Also X^y/y is increasing on $[2\nu,2\delta]$, which gives the factor $1/\max(2\delta,|\Delta|)$. The crowding budget at

$s_0 = 1+\eta+i\gamma_r$ bounds the total cost of all representatives near γ_r. The maximum of the linear functional under this budget is the fractional knapsack value, computed greedily. The repulsion theorem applies because all representatives counted in the cell (k,l) have $1-\beta \le \delta_k$ and height $\le e^l$; for half-integer l we use the integer cell above. □

Lemma 6.4 (crowding). *For a window of height ω centred at $t \le T$ and any $\eta > 0$, the number of representatives in the window is at most $n(\omega) = \min\big(\lfloor M'/\min(f_\omega(\eta), f_\omega(\eta+\delta))\rfloor, \lceil\omega/\delta\rceil+1\big)$, where $f_\omega(a) = a/(a^2+\omega^2/4)$ and $M' = \frac{1}{2}\log\frac{T}{2\pi} + \frac{1}{\eta} - 0.36$. Consequently Bellotti's c_4 [4, (4.13)–(4.14)] may be computed with $I' = \delta\sum_i n(\omega)\max_{[y_i,y_i+\omega]} g_\alpha$ in place of $I = \int_1^T g$, where $g_\alpha(y) = e^{y/6}y^{1.5-2\alpha}e^{-\pi y/2}$, $\omega = 1/4$, and the windows tile $[1,60]$.*

Proof. This is the crowding argument above applied to $s = 1+\eta+it$. For $0 < \eta \le 1.5$ we have $-\zeta'(1+\eta)/\zeta(1+\eta) \le 1/\eta - 0.36$; the minimum of $1/\eta + \zeta'/\zeta(1+\eta)$ on $(0,1.5]$ is 0.378. Beyond $y = 60$ the function g_α is below 10^{-30}. The worst case is taken over each δ-bin, and η is optimised on a grid. □

6.3. **The Barban–Vehov mean square.** Let $\Delta = \log(V/U)$, $Y_1 = \log(X/V)$ and $Y_2 = \log(X/U)$.

Theorem 6.5. *Let $\log U \ge 40$, $X > V$, $\tau \in [1.3, 2.5]$ and $\log X/\log V \in [1.1, 2.6]$. Then*

$$S := \sum_n \frac{\Psi(n)^2}{n} e^{-n/X} \le e^{-1/X} + D_4 + \mathrm{REM} + 2\sqrt{S_s S_t} + S_t,$$

where

$$D_4 = \Delta^{-2}\Big[\frac{Y_2^2 - Y_1^2}{2} - \gamma_E\Delta + k_1(Y_1+Y_2-2\gamma_E) + 2k_0\Big], \qquad k_1 \le -0.5853, \quad k_0 \le 0.4214.$$

Here REM *is an explicit sum of exponentially small contour remainders, S_t is an explicit bound for the mean square of the non-smooth part of the weights, and $S_s = e^{-1/X} + D_4 + \mathrm{REM}$.*

Outline. Write $\Psi = (\Lambda_V - \Lambda_U)/\Delta$, where $\Lambda_z(n) = \sum_{d\mid n}\mu(d)\log^+(z/d)$. Split each truncation weight $\rho(d)$ as $\rho_s + \rho_t$, where $\rho_s(d) = \mathbb{E}[\rho(de^{-\eta Z})]$ with $Z \sim N(0,1)$ and $\eta = 0.02$. By Minkowski's inequality in $\ell^2(e^{-n/X}/n)$, $S \le (\sqrt{S_s} + \sqrt{S_t})^2$.

For the smooth part, the Mellin transform of ρ_s is $\hat\rho(s)e^{\eta^2 s^2/2}$, and

$$S_s = (2\pi i)^{-3}\iiint \Gamma(w)X^w\hat\rho_s(s_1)\hat\rho_s(s_2) \times \frac{\zeta(1+w)\zeta(1+w+s_1+s_2)}{\zeta(1+w+s_1)\zeta(1+w+s_2)}\mathcal{A}(w,s_1,s_2)\,dw\,ds_1\,ds_2,$$

where $\mathcal{A}$ is an Euler product that converges absolutely near 0 and satisfies $\mathcal{A}(w,0,s_2) = \mathcal{A}(w,s_1,0) = 1$.

The Gaussian factor allows every contour to be closed inside the box $|\operatorname{Im}| \le T_2 = 400$, $\operatorname{Re} \ge 1-\theta$, $\theta = 0.4$. In this box $1/\zeta$ is analytic; this follows from [24] and is in any case confirmed by the computation. Beyond the box all lines lie at $\operatorname{Re} \ge 1$, and the Gaussian factor gives $e^{-\eta^2 T_2^2/2}$.

The residues produce $e^{-1/X} + D_4$, with $k_1 = H(0)$ and $k_0 = H'(0) + \gamma_E H(0)$, where H is the kernel of the s-integral at $w = 0$. At leading order this reproduces the asymptotic $\frac{3}{2} + (\Theta - \tau)/(\tau - 1)$ of the Graham-type mean square.

The certified inputs are:

- rigorous box bounds $|\zeta| \le 11.89$ (for $|s-1| \ge 0.1$) and $|1/\zeta| \le 14.57$ on $\sigma \in [0.6, 1.3]$, $|t| \le 450$, obtained from Taylor models of $(s-1)\zeta(s)$ in `arb` (`lemG/box2.py`);

- enclosures $k_1 = -0.607063 \pm 0.0217$ and $k_0 = 0.35026 \pm 0.0711$ (`lemG/kcert2.py`);
- sup bounds for $\mathcal{A}$, obtained from enclosures over cells of the 3-torus of phases for $p \le 100$, a second-order bound for $p \le 10^6$, and a tail estimate beyond (`lemG/remainder.py`);
- S_t, for which $|\rho_t(d)| \le (\eta/\Delta)\big[\varpi((\log d - \log U)/\eta) + \varpi((\log d - \log V)/\eta)\big]$ with $\varpi(y) = \phi(y) - |y|\bar{\Phi}(|y|)$. The resulting block sum uses exact masses for $m \le 10^6$ and the bound $|\sum_{m \le x} \mu^2(m)/\varphi(m) - \log x - 1.3326| \le 4/\sqrt{x}$ [25] beyond (`lemG/stbound.py`).

The integrals of explicit majorants are evaluated as rigorous upper sums or in closed form. At the binding cell, REM $= 5.4 \cdot 10^{-4}$ and $S_t = 9.9 \cdot 10^{-5}$. All cells of the final tables have $\tau \in [1.54, 1.78]$, $\log X/\log V \in [1.42, 1.55]$ and $\log U \in [56.6, 78.6]$, and every one of them takes its d_4 from Theorem 6.5. Details are in `notes/LEMMA_G.md`, Step 3. □

Remark 6.6. At the binding cell, Theorem 6.5 gives $d_4 \approx 1.48$. At the same parameters, an X^ε majorant combined with the Ramaré–Zuniga Alterman bound [26] gives ≈ 20, and our earlier explicit Selberg-diagonalisation bound (`notes/LEMMA_F.md`) gives ≈ 5.8; Bellotti's Table 5 values, at her parameters, are 85–105. Since $\log N$ has elasticity about 1 in d_4, this is the single largest improvement in the density estimate.

7. The computation

7.1. **The scan.** For each interval $[L_1, L_2]$ we search for kernel parameters (c, f): first over the grid $c \in \{40, 50, \dots, 100\}$, $f \in \{0.5, \dots, 0.9\}$, then locally over $c \pm 10$ in steps of 5 and $f \pm 0.1$ in steps of 0.05. Given (c, f), we put $\tau_{12} = \varepsilon_R(L_2 + 1)/(1 + 2/L_1)$ and choose ε so that the smoothing term of Proposition 3.6 equals $f\tau_{12}$. This uses three fixed-point steps; the residual excess is bounded rigorously in `certify/transfer_check.py` and is below 10^{-3} relative even at $L_1 = 43$. We then evaluate the right-hand side of Proposition 3.6 with N^+ from §6, λ^* from Theorem 4.10 and $H = 3 \cdot 10^{12}$. The ratio q is this right-hand side divided by τ_{12}; an interval is certified when $q < 1$.

The scan starts with intervals of width 10 and bisects any interval that is not certified, down to width 0.5 (`code/interval_adapt.py`). For $[43, 57)$ we only need some bound, since this range enters only through the integral in Lemma 2.4; there we use widths 2 to 0.5, and q may exceed 1.

range of L	worst q	remark
$[43, 57)$	3.80 (at 43), ≤ 0.47 above 48	used only through Lemma 2.4
$[57, 1000)$	0.9966 at 907	$[857, 907)$ needed width 5
$[1000, 2000)$	0.8884 at 1000	
$[2000, 2810)$	0.4063	
$[2800, 3159)$	0.4001	

The binding range $L \approx 900$ corresponds to bandwidth $T_1 \approx e^{30}$. There the zero sum is dominated by $\lambda = (1-\beta)\log T \in [0.20, 0.32]$ and $\log T \in (28.73, 30]$. For the interval $[900, 910]$ of width 10, the best kernel gives 0.40 (smoothing) +0.60 (zeros above H_0) $= 1.002$, plus negligible terms; intervals of width 5 are certified.

7.2. **Certificates.** The following list gives, for each ingredient, the program that certifies it. All these programs are in the repository; `code/rerun_all.sh` re-runs the whole chain.

(1) Lemma 2.2 (ε_R on every cell): `certify/epsR_cert.py`.
(2) Lemma 3.1 (smoothing constant): `certify/smoothing_const.py`.
(3) Proposition 2.5 (from ψ to $\pi-$li, both conditions of (2.2)): `certify/transfer_check.py`. The worst ratio is 0.99577, at $\log y = 917$.

(4) Lemmas 4.2–4.9 (constants of [5]): in `rep/`, the programs `consts_cert.py`, `lemma5_cert.py`, `lemma11_cert.py`, `f_nonneg_cert.py`, `mp_cert.py` and `misc_cert.py`.
(5) Theorem 4.10: `rep/lamcert.py`, with output `rep/lamstar2c.json`.
(6) Theorem 5.2: `rep/repcert_run.py` and `rep/min_fix.py`; output `rep/dstar2c_all.json`.
(7) Theorem 6.5: in `lemG/`, the programs `box2.py`, `kcert2.py`, `remainder.py` and `stbound.py`.
(8) Theorem 6.1: the tables `lemmaD_tableGH0.01_*.pkl`, produced by `bt4.py`. Every cell's bound is recomputed exactly from its stored parameters when it is used.
(9) The scan: `interval_adapt.py`, with logs in `logs/`.

7.3. **Rounding in the scan.** Items 1–7 are certified in ball arithmetic. The scan (item 9) and the evaluation of the density bounds from the stored parameters (item 8) run in IEEE double precision, with ε_R and $-\zeta'/\zeta(1+\eta)$ taken from 60- and 30-digit `mpmath` computations. This is the one place where the argument rests on a floating-point computation rather than on a certificate, so we describe it precisely.

Each value of q is a sum of a few thousand non-negative terms. Each term is a product of exponentials and explicit factors, obtained from the parameters by a bounded number (well under 10^4) of floating-point operations. Only two subtractions involve quantities of comparable size. The first is $c_1 = e^{-L^2/U} - B_1 - \cdots$; over all table cells $c_1 \geq 0.92$, so there is no cancellation. The second is $c_1^2 - c_5$, which is ≥ 0.83 over all table cells, so it amplifies relative errors by at most a factor of about 1.2. The greedy knapsack value of Lemma 6.3 and the Stieltjes and upper sums of Proposition 3.6 are sums of non-negative terms. Standard error analysis therefore bounds the relative rounding error of each q by about 10^{-11}.

As an empirical check, inflating every density bound by a factor $1 + 10^{-6}$ changes q on the binding interval $[907, 912]$ by $5.8 \cdot 10^{-7}$. This is consistent with the scan depending on the density bounds linearly, with coefficient at most 1. The smallest margins are $1 - q \geq 3 \cdot 10^{-3}$ in the scan and about $4 \cdot 10^{-3}$ in Proposition 2.5, some eight orders of magnitude above the rounding error. A complete interval-arithmetic re-evaluation of the scan, keeping the floating-point parameters but evaluating each bound in `arb`, is straightforward but costly; we have not carried it out.

7.4. **Adversarial audits.** Two adversarial audits were carried out by separate instances of the same AI system that assisted with the computations (see the acknowledgements); they are not independent in the human sense. The first audit covered the new ingredients (§§4–6) and checked the following points, confirming them:

- the legitimacy of keeping the second pair in Lemma 6 of [5];
- the per-σ form of Lemma 5;
- the induction in Theorem 4.10;
- the use of d^* in the pair sum;
- the error constants;
- the replacement of $\mathcal{J}$.

It found one genuine defect: the clamp on ν described at the end of §5. It also found three minor gaps: a missing non-negativity certificate for the dropped Lemma-10 terms, the unrounded w-derivative values, and the range of the Hiary–Patel–Yang bound. All four were fixed, and the full chain was re-run with identical results.

A second audit, of a draft of this paper, found an unjustified rescaling: the density constants, which are valid at the corner δ_k of a cell, had been applied with the actual δ inside the cell. This was removed (Theorem 6.1 now uses the corner value), and the scan was re-run. That audit also corrected the definition of N_R, the attributions for the lower range, the hypotheses of Lemmas 4.3 and 2.2, the statement of the correction factor ι, and the reproducibility scripts.

8. Remarks

8.1. **Corrections to inputs.** For convenience we collect the points found while re-deriving the explicit inputs; none of them changes a published main theorem.

(1) In [5]:
   (a) the Lemma-10 term in the final inequality should carry the factor κ (Remark 4.1);
   (b) the misprint in (25) (Remark 4.1);
   (c) the value of m_5 in Table 1 (Remark 4.8);
   (d) the integral of the digamma majorant (31) (§4.2);
   (e) the uniform test for (21), which needs $\sigma \geq 0.993464$ (Remark 4.4);
   (f) the numerical verification of $\partial_\sigma G_p < 0$ is unnecessary, since this follows from $P \geq 0$ (Lemma 4.7);
   (g) the terms with $m > 15$ in Lemma 10 are dropped there without a bound, which is justified by $f \geq 0$.

(2) In Büthe's smoothing estimate, the quoted asymptotic for ν_c does not match the stated normalisation (Remark 3.2).

(3) In Bellotti [4], the pair-sum accounting [4, (4.11)] assumes a spacing of δ between representatives of different rectangles, which does not appear to be justified as stated; Lemma 6.3 avoids this assumption. Since her constants are much larger than ours elsewhere, her published bounds are not affected in practice.

8.2. **On two recent preprints.**

(1) *arXiv:2408.02591* [11] claims (1.1) for $x \geq e^{43.5102147}$. The argument combines $\pi(x) < e\,\pi(x/e)$ with $\pi(x) > x/\log x$ to conclude, in (29), that $\pi(x)^2 - \frac{ex}{\log x}\pi(x/e) < 0$. That conclusion requires $\frac{x}{\log x} e\pi(x/e) \geq \pi(x)^2$. From the first inequality one only obtains $\frac{x}{\log x} e\pi(x/e) > \frac{x}{\log x}\pi(x)$. Passing from there to $\pi(x)^2$ would need $\pi(x) \leq x/\log x$, which is the reverse of the second inequality (and is false for $x \geq 17$). The gap is not a slip that can be repaired: the difference of the two sides of (1.1) has size $x^2/\log^6 x$, whereas the bounds used have relative accuracy $\approx (\log x)^{-3/2}$.

(2) *arXiv:2407.12052* [10] claims (1.1) for $x \geq e^{547}$. The argument has four problems.
   (a) The monotonicity argument assumes Sterneck's conjecture $|M(x)| < \frac{1}{2}\sqrt{x}$, which is false [21].
   (b) It asserts $M(x) = O(\sqrt{x})$, which is an open problem (and widely believed false); the proof given yields at most $O(x^{1/2+\varepsilon})$.
   (c) The retained main term is itself only $O(x^2/\log^2 x)$, so the sign of $G(x) = \pi(x)^2 - \frac{ex}{\log x}\pi(x/e)$ is not determined.
   (d) The range below $e^{3158.442}$ is "covered" by evaluating at 26 sample points.

Neither preprint therefore proves (1.1) in the range considered here.

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